# Group I cross section (one gluon in the final state): check against `Calculation_main.tex`

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## 1 Answer

**How it was checked.** The same way as Groups II and III. Every displayed equation of `Group_I.tex` was transcribed into code and compared with a first-principles evaluation of the same quantity.

- **Reference.** Eq. (CS1) of `Calculation_main.tex` with the coefficients of the same file: $\mathbb{A}^{(1)}$, $\mathbb{A}^{(3)}$, $\mathbb{B}_3$, $\mathbb{B}_2$ and $\mathbb{C}_1$ (Sec. 2). The entries of $\mathbb{B}_3$ that had not been re-derived before (the instantaneous one and the two two-$\rho$ ones) were rebuilt here from the matrix elements m1 and m5.
- **Evaluation.** The comparison is made at random transverse points, with random SU(3) adjoint Wilson lines and random charges. Each comparison is done twice: with your convention $U^{bc} = U^{cb}$ and with a general $U$.
- **Integrated lines.** Lines after a $p^+$ or $w$ integration are compared with the exact numerical integral of the (already verified) line before it. Where the integrand has a pole inside the range, this is the principal value.
- **No limits.** No limit is taken in $\vee$, $k^+$, $p^+$ or $\Lambda$. The only conditions used are $\vee > p^+, k^+ > \Lambda$ and the support of the coefficients.

Equation numbers (1.1)–(1.30) are those of your compiled `Group_I.tex`. The corrected file is `Group_I_corrected.tex`, with changes in blue and removed text in red.

**Correct as written.** Everything below holds with $U^{bc} = U^{cb}$, and also for a general $U$: your index placements in Group I are the general ones.

- **Row II.**
  - Part 1, Eq. (1.1): prefactor $\frac{g^4 N_c}{32\pi^4 k^+}$, colour structure and kernel. The rewriting (1.2) is also right.
  - The bracket $[-\frac{2}{\epsilon} - 2\gamma - \log\frac{(x-w)^2\mu^2}{4}]$ equals $\frac{1}{\pi}\int_z [K(x,w,z) - K(w,w,z)]$ exactly, in $d_\perp = 2-\epsilon$ and $\overline{\text{MS}}$. This is the scheme of Row V part 2b, so the poles of the two rows cancel (Sec. 9).
  - Part 2, Eq. (1.3): prefactor $-\frac{g^4}{64\pi^6 k^+}$.

  - The split into 2a and 2b, Eqs. (1.4)–(1.9): $\mathfrak{F}$, its closed form and $R^i$. This assumes the bracket contains $-k^i \log \frac{\vee}{k^+}$, which (1.3) omits (G2).

- **Row III.**
  - The explicit form (1.10) is exactly Eq. (CS1) with $\mathbb{B}_3$ *as written* in `Calculation_main.tex`.
  - Every $p^+$ integration is exact: all lines of $N_1$, $N_2$, $N_{3a}$, $N_{3b}$, Eqs. (1.11)–(1.14), including the arctan function $\mathcal{A}$. They agree with the numerical integrals to $10^{-14}$, provided each logarithm is read as $\log|\cdot|$ (G4).
  - The two red lines of $N_2$ cancel each other, as you indicated.

- **Row IV: correct as written**, Eqs. (1.15)–(1.16).
  - The explicit form, including $\mathbb{B}_2^{dc}$ (your index order is right).
  - The $p^+$ integration over $\Lambda < p^+ < \vee - k^+$. It is exact, with no limit taken.

- **Row V.**
  - Part 1 (new here), Eqs. (1.17)–(1.19): the explicit form, the $w$ integration, and the coefficients and phases of the two endpoint terms. These coefficients multiply the exact $\log \frac{k^+-\Lambda}{\Lambda}$.
  - Part 2, Eqs. (1.20)–(1.29): as checked before (`RowV_UV.pdf`, `Part2b_MSbar.pdf`); the explicit form (1.20) was checked again here.
  - The plus prescription of $d^3N_{2b\,\text{finite}}$ is in `PlusPrescription_2b.pdf`.

**Mistakes.**

**G1 Row III: the $p^+ > k^+$ non-instantaneous term has the wrong sign** (Sec. 3).

- This is the fourth line of (1.10) and, consequently, every line of $N_2$ (1.12).
- It comes from $\mathbb{B}_3$ in `Calculation_main.tex` (correction E3). The entry $\Theta(p^+ - k^+ - \Lambda)$ of $\mathbb{B}_3$ is $-(2\pi)^3$ times the light-cone wave function; every other entry is $+(2\pi)^3$ times it.
- With the sign as written, Row III keeps a term $\propto \frac{1}{\Lambda}$. With the corrected sign, all the green terms ($\frac{1}{\Lambda}$, $\frac{1}{\vee - k^+}$, $\frac{1}{k^+ - \Lambda}$) cancel exactly.

**G2 Row II, part 2: the bracket of (1.3) lacks $-k^i \log \frac{\vee}{k^+}$** (Sec. 4).

- The term is in $\mathbb{A}_2^{(3)}$ of `Calculation_main.tex`.
- Your split (1.4)–(1.9) already assumes it: $d^3N_{2a} + d^3N_{2b}$ equals (1.3) *with* this term.
- In (1.3) and (1.8) the index $a$ appears three times. The colour factor must read $U^{ah}(w) f^{hbc}\{\rho^b, \rho^c\} - f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d, \rho^e\}$, as you wrote it in (1.5).

**G3 Row II: the commutator $[\mathbb{N}_2, \mathbb{A}^{(1)}]$ is missing** (Sec. 5).

- `Calculation_main.tex` sets $\mathbb{N}^{(2)}\mathbb{A}^{(1)} = \mathbb{A}^{(1)}\mathbb{N}^{(2)}$ (correction E5). This changes only the three-$\rho$ term $\propto \log \frac{\vee}{\Lambda}$ of part 2a.
- In the $U^{ah}(w)$ term, $\log \frac{\vee}{\Lambda}\,\mathfrak{F} \to \log \frac{\vee}{\Lambda}\,\mathfrak{F}_\beta$. This is fix F4 of the earlier three-$\rho$ and BFKL notes.

**G4 Row III: principal values** (Sec. 6).

- Every logarithm in $N_1$, $N_2$ and $N_{3a}$ whose argument contains $k^+(x-w)^2 - c\,(x-z)^2$ must be $\log|\cdot|$.
- The integrand has a pole at $p_*^+ = k^+(x-w)^2/(x-z)^2$. When $p_*^+$ lies in the integration range, the argument of the logarithm is negative; the principal value then gives the modulus. Your Fourier transforms (i3)–(i5) are principal values too.

**G5 Limits that are not exact** (Sec. 7). You do not want limits in $\Lambda$, $\vee$, $k^+$; these steps take one:

(a) The bracket of Row II part 1 (1.1), (1.2) is the $\Lambda \to 0$ form. Sec. 7 gives the exact one.

(b) The bracket of Row II part 2 (1.3) is the $\Lambda \to 0$ form. Sec. 7 gives the exact one and the exact $R^i$; $\mathfrak{F}$ (part 2a) does not change.

(c) The plus prescription in Row V must keep the limits $\Lambda < p^+ < k^+ - \Lambda$, both in part 1 (1.19) and in part 2a (1.26). With $\int_0^{k^+}$ the result is off by $O(\Lambda/k^+)$: in the tests, $2\times10^{-5}$ to $5\times10^{-1}$ for part 1 and $2\times10^{-4}$ to $4\times10^{-1}$ for 2a. With the limits kept, it is exact to $10^{-14}$. This is the same point as M2 of Groups II and III.

(d) In (1.28), $4\frac{\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}$ is dropped but $2\log(1 - \frac{\Lambda}{k^+})$ is kept. Keep (1.27), which is exact.

**G6 Notation** (Sec. 8):

- $\log[1-\frac{k^+}{\vee}]^2$ in (1.1), (1.2) is $\log^2(1-\frac{k^+}{\vee})$;
- $\int_{x'}$ is missing in (1.3) and (1.8);
- $S^{ad}\to U^{ad}$ in the identity before (1.23);
- the phase $e^{-ik\cdot(z-x)p^+/k^+}$ belongs inside $\int dp^+$ in (1.18), (1.21)–(1.25);
- there is a missing comma in $\int_{x,z,y,x',y'w'}$.

**Not in `Group_I.tex`.** `Group_I.tex` does not contain:

- Row I, which has only four-$\rho$ terms;
- the four-$\rho$ parts of Rows III and IV.

For your four-$\rho$ list, Sec. 10 gives the latter two, checked against first principles.

## 2 Reference

**Cross section.** Eq. (CS1):

$$\frac{d^3N^{(\mathrm{I})}}{d^3k}=\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik\cdot(w'-w)}\,\mathcal{C}_i^a(w')\,\mathcal{V}_i^a(w)+\text{c.c.},\qquad \mathcal{C}_i^a=-\bar{\mathbb{A}}_i^{\dagger a}(k^+,-w')+U^{ab'}(w')\mathbb{A}_i^{\dagger b'}(k^+,-w'). \tag{2.1}$$

$\mathcal{C}$ is the conjugate LO amplitude:

$$\mathcal{C}_i^a=\frac{ig}{\sqrt{2}\pi\sqrt{k^+}}\int_{x'}\mathcal{K}^i(x'-w')\big[U(x')-U(w')\big]^{ab'}\rho^{b'}(x'),\qquad \mathcal{K}(X)=\frac{X}{X^2}. \tag{2.2}$$

The rows are

$$\text{II: }\mathcal{V}=U^{ab}(w)\mathbb{A}_i^{(3)b}(k^+,-w)-\bar{\mathbb{A}}_i^{(3)a}(k^+,-w), \tag{2.3}$$

$$\text{III: }\mathcal{V}=\int\frac{dp^+}{2\pi}\int_z\bar{\mathbb{B}}_{3ji}^{\dagger ba}(k^+,w;p^+,-z)\big[-\bar{\mathbb{A}}_j^b(p^+,-z)+U^{bc}(z)\mathbb{A}_j^c(p^+,-z)\big], \tag{2.4}$$

$$\text{IV: }\mathcal{V}=2\int\frac{dp^+}{2\pi}\int_z\big[U^{ac}(w)U^{bd}(z)\bar{\mathbb{A}}_j^{\dagger b}(p^+,-z)\mathbb{B}_{2ji}^{dc}(k^+,-w;p^+,-z)-\bar{\mathbb{A}}_j^{\dagger b}(p^+,-z)\bar{\mathbb{B}}_{2ji}^{ba}(k^+,-w;p^+,-z)\big], \tag{2.5}$$

$$\text{V: }\mathcal{V}=2\int\frac{dp^+}{2\pi}\int_{x,z}\big[-\bar{\mathbb{C}}_{1jki}^{\dagger bca}\bar{\mathbb{B}}_{2jk}^{bc}(k^+-p^+,-z;p^+,-x)+\bar{\mathbb{C}}_{1jki}^{\dagger dca}U^{db}(x)U^{ce}(z)\mathbb{B}_{2jk}^{be}(k^+-p^+,-z;p^+,-x)\big], \tag{2.6}$$

with $\bar{\mathbb{C}}_1=\mathbb{C}_1(k^+,w-z;p^+,z-x)$ and $\mathbb{A}_i^a(P,-W)=\frac{ig}{\sqrt{2}\pi\sqrt{P}}\int_X\mathcal{K}^i(X-W)\rho^a(X)$.

- A bar means $\rho\to U\rho$, and † the complex conjugate of the coefficient.
- In $\mathbb{B}_{2ji}^{bc}(k^+,-w;p^+,-z)$ the gluon $(c,i,k^+)$ is at $w$ and the gluon $(b,j,p^+)$ at $z$.
- In $\mathbb{B}_{3ji}^{ba}(k^+,w;p^+,-z)$ the gluon $(a,i,k^+)$ is at $w$ and the gluon $(b,j,p^+)$ at $z$.

**$\mathbb{B}_3$.** All four entries of $\mathbb{B}_3$ in momentum space were compared with $(2\pi)^3$ times the one-gluon wave functions, rebuilt from the matrix elements (Table 1).

- The instantaneous entry (m5) and the two two-$\rho$ entries, $\{\rho,\rho\}$ and $[\rho,\rho]$ (m1 twice), are new here.
- For the two-$\rho$ entries, the two time orderings give $c_1\rho^b(-\mathbf{p})\rho^a(\mathbf{k})+c_2\rho^a(\mathbf{k})\rho^b(-\mathbf{p})$, which is $\frac{c_1+c_2}{2}\{\rho^a,\rho^b\}-\frac{c_1-c_2}{2}if^{abc}\rho^c(\mathbf{k}-\mathbf{p})$.
- The non-instantaneous entries were checked in `groups23/main` (m2, m3).

| entry of $\mathbb{B}_3$ | $p^+<k^+$ | $p^+>k^+$ |
|---|---|---|
| instantaneous (m5), new | +248.050213 | +248.050213 |
| two-$\rho$, $\{\rho,\rho\}$ (m1, m1), new | +248.050213 | +248.050213 |
| two-$\rho$, $[\rho,\rho]$ (m1, m1), new | +248.050213 | +248.050213 |
| non-instantaneous (m2, m3), `groups23/main` | +248.050213 | **−248.050213** (E3) |

Table 1: $\mathbb{B}_3$ divided by the wave function ($8\pi^3=248.0502$). The spread over random momenta is below $10^{-12}$.

The coordinate forms of all entries follow from the momentum forms through your Fourier transforms (i2)–(i5). Each was redone by hand and agrees with `Calculation_main.tex`. In (i4) the index of $(\mathbf{k}-\mathbf{p})$ ends up on $\frac{X+\xi Y}{X^2-\xi Y^2}$; this is why the vertex bracket of $\mathbb{B}_3$ reads $\delta_{ik}\delta_{jm}-\frac{k^+}{p^+}\delta_{jk}\delta_{im}-\ldots$, not $\delta_{im}\delta_{jk}-\ldots$ as in $\mathbb{C}_1$.

# 3 G1: the sign of the $p^+ > k^+$ non-instantaneous term of Row III

**The correction.** The non-instantaneous entry of $\mathbb{B}_3$ for $p^+ > k^+ + \Lambda$ needs a minus sign, in momentum space and in coordinate space:

$$\Big[+-\frac{1}{((x-z)-(x-u))^2}\,\Theta(p^+-k^+-\Lambda)-\frac{p^+k^+}{(k^+(x-z)-p^+(x-u))^2}\,\Theta(k^+-\Lambda-p^+)\Big]. \tag{3.1}$$

In Row III this changes the sign of the fourth line of (1.10):

$$\begin{aligned}&+-\frac{ig^4f^{abd}}{4\pi^4}\frac{1}{k^+}\int_{\Lambda+k^+}^{\vee}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{(x'-w')^i(y-z)^j(z-w)^k}{(x'-w')^2(y-z)^2(z-w)^2}\,\frac{k^+(x-w)^m-p^+(x-z)^m}{k^+(x-w)^2-p^+(x-z)^2}\,\frac{1}{k^+-p^+}\\&\times\Big[\delta_{ik}\delta_{jm}-\frac{k^+}{p^+}\delta_{jk}\delta_{im}-\frac{k^+}{k^+-p^+}\delta_{ij}\delta_{km}\Big]\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')\big[U^{de}(x)U^{bc}(z)-U^{de}(x)U^{bc}(y)\big]\rho^e(x)\rho^c(y),\end{aligned} \tag{3.2}$$

and so the sign of every line of $N_2$:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k}\Big|_{(1.12)}\;\to\;-\,\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k}\Big|_{(1.12)}. \tag{3.3}$$

Your $p^+$ integration of this line is right (Table 3). Only the sign of the integrand changes.

**Why.** There are three independent reasons.

1. **Light-cone wave function.** For $p^+ > k^+$ the valence charge first adds a gluon (m2), and the three-gluon vertex then merges the pair (m3). Rebuilt this way, the wave function is $-\frac{1}{(2\pi)^3}$ times the entry of $\mathbb{B}_3$ as written. For $p^+ < k^+$, and for every other entry, the factor is $+\frac{1}{(2\pi)^3}$ (Table 1).

2. **The green terms.** The terms of Row III with $\frac{1}{\Lambda}$, $\frac{1}{\vee-k^+}$ and $\frac{1}{k^+-\Lambda}$ are the first lines of $N_1$ (two of them), $N_2$ and $N_{3a}$. With $A\equiv(x-z)^2$, $B\equiv(x-w)^2$, $\mathcal{K}_1\equiv\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}$ and the common colour factor, they are

$$\frac{ig^4f^{abd}}{8\pi^5}\,\frac{\mathcal{K}_1}{A-B}\Big[\Big(\frac{1}{\Lambda}-\frac{1}{\vee-k^+}\Big)+\Big(\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\Big)\pm\Big(\frac{1}{\Lambda}-\frac{1}{\vee-k^+}\Big)-\Big(\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\Big)\Big]. \tag{3.4}$$

   The upper sign is as written; it leaves $2\frac{ig^4f^{abd}}{8\pi^5}\frac{\mathcal{K}_1}{A-B}(\frac{1}{\Lambda}-\frac{1}{\vee-k^+})$, which diverges as $\Lambda\to0$. With the lower sign (G1) the sum is zero. Numerically the residual is below $10^{-15}$ relative to the uncorrected sum.

3. **The $\log(\vee/k^+)$ terms.** Without G1, the coefficient of $\log\vee$ does not match BFKL⊗LO. This was fix P1 of the BFKL note: the large-$p^+$ limit of this line must be $+2K_h$, not $-2K_h$.

# 4 G2: Row II, part 2

In $\mathbb{A}_2^{(3)}$ of `Calculation_main.tex`, coordinate form, the two-$\rho$ bracket ends with $-k^i\log\frac{\vee}{k^+}$. Eq. (1.3) should read

$$\begin{aligned}\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k}\;=\;&-\frac{1}{(2\pi)^3}\frac{g^4}{64\pi^6}\frac{1}{k^+}\int_{w,w'}e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y,x'}e^{ik\cdot(w-y)}e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2p^2k^2}\Big[\cdots-k^i\log\frac{\vee}{k^+}\Big]\\&\times\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')\Big[U^{ah}(w)f^{hbc}\{\rho^b(x),\rho^c(y)\}-f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\}\Big],\end{aligned} \tag{4.1}$$

where $[\cdots]$ is your bracket. With $E^i=\frac{k^2p^i-(k\cdot p)k^i}{(k-p)^2}$, $C^i=\frac{(2p^2-2k\cdot p+k^2)k^i+k^2p^i}{(k-p)^2}=k^i+D^i$ and $D^i=\frac{p^2k^i+k^2p^i}{(k-p)^2}$:

- the full bracket equals $(E+2k+D)^i\big[\log\frac{\vee}{k^+}-\log\frac{\vee}{\Lambda}\big]+R^i$ exactly, with your $R^i$ (1.9). Without the extra term the coefficient of $\log\frac{\vee}{k^+}$ would be $E+3k+D$;
- the transform of the odd part of $(E+2k+D)^i/(k^2p^2)$ is $-2\pi i\,\mathfrak{F}^i$, which gives your (1.5)–(1.7). The transform is done step by step in Part I of `RowII_FT_and_evolution.pdf`;
- numerically, (1.3) as written minus the bracket of `Calculation_main.tex` is exactly $k^i\log\frac{\vee}{k^+}$ (rel. $2\times10^{-15}$), and $2a+2b$ reproduces the full bracket (rel. $2\times10^{-15}$);
- the prefactor and colour factor of the corrected (1.3) agree with first principles (rel. $2\times10^{-14}$).

# 5 G3: Row II and the commutator $[\mathbb{N}_2, \mathbb{A}^{(1)}]$

`Calculation_main.tex` drops $-[\mathbb{N}^{(2)}, \mathbb{A}_2^{(1)}]$ from the unitarity relation for $\mathbb{A}_1^{(3)\dagger}$ (correction E5). These two operators do not commute: $[\int_{x,y} K(x,y)\rho^b(x)\rho^b(y), \rho^a(u)] = if^{bac}\int_x K(x,u)\{\rho^b(x),\rho^c(u)\}$, with $K(x,y) = \int_z K(x,y,z)$.

- The $-2k^i \log\frac{\vee}{\Lambda}$ term of the Row II bracket is exactly $\log\frac{\vee}{\Lambda}[n, \mathbb{A}_2^{(1)}]$, where $\mathbb{N}^{(2)} = n\log\frac{\vee}{\Lambda}$.
- In coordinate space it is the $\mathfrak{F}_\alpha$ part of $\mathfrak{F}$.
- The $p^+$ ordering of the virtual factor puts the gluons softer than $k^+$ to the left of $\mathbb{A}^{(1)}$ and the harder ones to the right. With this ordering the $\log\frac{\vee}{\Lambda}$ part of the commutator is removed from the unrotated term.

Part 2a then reads

$$\begin{aligned}\frac{d^3N_{2a}}{d^3k} = &-\frac{1}{(2\pi)^3}\frac{ig^4}{32\pi^5}\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x'}\frac{(x'-w')^i}{(x'-w')^2}\big[U^{ab'}(x') - U^{ab'}(w')\big]\rho^{b'}(x')\\ &\times\Big\{\Big[\log\frac{\vee}{\Lambda}\,\mathfrak{F}^i_\beta - \log\frac{\vee}{k^+}\,\mathfrak{F}^i\Big]U^{ah}(w)f^{hbc}\{\rho^b(x),\rho^c(y)\}\\ &-\Big[\log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+}\Big]\mathfrak{F}^i\, f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\}\Big\},\end{aligned} \tag{5.1}$$

with $\mathfrak{F} = \mathfrak{F}_\beta + \mathfrak{F}_\alpha$ split as in your (1.6):

$$\mathfrak{F}^i_\beta = \frac{(x-w)^i}{(x-w)^2}\int_z\frac{(w-z)\cdot(y-z)}{(w-z)^2(y-z)^2} - \frac{(y-w)^i}{(y-w)^2}\int_z\frac{(w-z)\cdot(x-z)}{(w-z)^2(x-z)^2}, \tag{5.2}$$

$$\mathfrak{F}^i_\alpha = -\Big[\frac{(x-w)^i}{(x-w)^2} - \frac{(y-w)^i}{(y-w)^2}\Big]\int_z\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}. \tag{5.3}$$

Note the following:

- $\mathfrak{F}_\alpha$ and $\mathfrak{F}_\beta$ are separately logarithmically divergent at large $z$, like $\mathbb{N}_2$; the divergent parts are proportional to the total charge.
- The change is a three-$\rho$ term $\propto \log\frac{\vee}{\Lambda}$. It was established earlier: the symmetrized three-$\rho$ terms cancel only with it (fix F4), the two-$\rho$ terms then give JIMWLK⊗LO, and the coefficient of $\log\vee$ gives BFKL⊗LO.
- $\mathbb{N}_2$ in Row I multiplies four charges, so Row I itself is unchanged; its reordering to symmetric charges belongs to the four-$\rho$ list.

# 6 G4: principal values in Row III

Every line of (1.10) contains $\frac{1}{p^+(x-z)^2 - k^+(x-w)^2}$. This has a pole at

$$p^+_* = k^+\frac{(x-w)^2}{(x-z)^2}, \tag{6.1}$$

which for some transverse points lies inside $(\Lambda, k^+-\Lambda)$, $(k^+ + \Lambda, \vee)$ or $(\Lambda, \vee)$. The $p^+$ integral is then a principal value, as are your Fourier transforms (i3)–(i5). Each logarithm whose argument changes sign must therefore be read as

$$\log\frac{\vee(x-z)^2 - k^+(x-w)^2}{(\Lambda+k^+)(x-z)^2 - k^+(x-w)^2} \;\to\; \log\left|\frac{\vee(x-z)^2 - k^+(x-w)^2}{(\Lambda+k^+)(x-z)^2 - k^+(x-w)^2}\right|, \tag{6.2}$$

and the same for the logarithm of $N_1$ with $(k^+-\Lambda, \Lambda)$, the one with $(\vee, \Lambda)$, the six of $N_2$ and the six of $N_{3a}$. With this reading, all lines agree with the numerical principal-value integrals, also when the pole is inside the range. Without it they do not (Table 3). The arctan function $\mathcal{A}$ and the logarithm of $N_{3b}$ have no such problem.

# 7 G5: exact forms (no limit in $\Lambda$)

**(a) Row II, part 1.** With $\delta = \Lambda/k^+$ and $\Lambda < k^+ < \vee - \Lambda$, the big bracket of (1.1) is exactly (correction X6 of `Main_corrections.pdf`; $P_\epsilon \equiv -\frac{2}{\epsilon} - 2\gamma - \log\frac{(x-w)^2\mu^2}{4}$):

$$P_\epsilon\Big[\frac{11}{3} + \log\frac{\Lambda^2}{k^{+2}-\Lambda^2} + \log\Big(1-\frac{k^+}{\vee}\Big) - 8\delta + 2\delta^2 - \frac{4}{3}\delta^3\Big] + \frac{\pi^2}{3} - \frac{67}{9} + \log^2\Big(1-\frac{k^+}{\vee}\Big) + \log\Big(1-\frac{k^+}{\vee}\Big)\log\frac{\vee}{k^+} + R_\delta,$$

$$R_\delta = \frac{46}{3}\delta - \frac{4}{3}\delta^2 + \frac{8}{9}\delta^3 + \Big(\frac{22}{3} - 8\delta + 2\delta^2 - \frac{4}{3}\delta^3\Big)\log(1-\delta) - \Big(8\delta - 2\delta^2 + \frac{4}{3}\delta^3\Big)\log\delta$$

$$-2\log\delta\log(1-\delta)-4\,\mathrm{Li}_2(\delta)-\log^2(1+\delta)+\log(1+\delta)\log\delta. \tag{7.1}$$

Your bracket is its $\delta\to 0$ limit. This follows from

$$\frac{11}{3}+\log\frac{\Lambda}{k^+}+\log\Big(\frac{\vee}{k^+}-1\Big)-\log\frac{\vee}{\Lambda}=\frac{11}{3}+2\log\frac{\Lambda}{k^+}+\log\Big(1-\frac{k^+}{\vee}\Big). \tag{7.2}$$

For $\vee-\Lambda<k^+<\vee$ only one of the two wave functions contributes; that form is in X6.

**(b) Row II, part 2.** Exactly (corrections X4, X7; $\Lambda<k^+<\vee-\Lambda$), the bracket of (1.3) is

$$\log\frac{\Lambda}{k^+-\Lambda}E^i-2k^i\log\frac{\vee}{\Lambda}-2k^i\log\frac{\Lambda k^2+k^+p^2}{\vee k^2+k^+p^2}+C^i\log\Big[1+\Big(\frac{\vee}{k^+}-1\Big)\frac{k^2}{(k-p)^2}\Big]-C^i\log\Big[1+\frac{\Lambda}{k^+}\frac{k^2}{(k-p)^2}\Big]$$
$$-D^i\log\frac{\vee-k^+}{\Lambda}-k^i\log\frac{\vee}{k^++\Lambda}. \tag{7.3}$$

It splits exactly into $(E+2k+D)^i[\log\frac{\vee}{k^+}-\log\frac{\vee}{\Lambda}]$, which gives your 2a unchanged, plus

$$\begin{aligned}R^i_{\text{exact}}=&-2k^i\log\frac{p^2+\frac{\Lambda}{k^+}k^2}{k^2+\frac{k^+}{\vee}p^2}+C^i\log\Big[\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\Big(1-\frac{k^2}{(k-p)^2}\Big)\Big]-D^i\log\Big(1-\frac{k^+}{\vee}\Big)\\&+E^i\log\frac{k^+}{k^+-\Lambda}-C^i\log\Big[1+\frac{\Lambda}{k^+}\frac{k^2}{(k-p)^2}\Big]+k^i\log\Big(1+\frac{\Lambda}{k^+}\Big)\end{aligned} \tag{7.4}$$

(checked numerically: rel. $2\times10^{-15}$).

**(c) Row V: plus prescription.** The $p^+$ range of Row V is $\Lambda<p^+<k^+-\Lambda$. For any smooth $f$, exactly,

$$\int_\Lambda^{k^+-\Lambda}dp^+\frac{k^+}{p^+}f(p^+)=\int_\Lambda^{k^+-\Lambda}dp^+\frac{k^+}{p^+}\big[f(p^+)-f(0)\big]+k^+f(0)\log\frac{k^+-\Lambda}{\Lambda}, \tag{7.5}$$

and the same at $p^+\to k^+$. Your endpoint terms are exactly the last term, with $\log\frac{k^+-\Lambda}{\Lambda}=\log\frac{\vee}{\Lambda}+\log\frac{k^+-\Lambda}{\vee}$. The remaining integrals in (1.19) and (1.26), however, run over $0<p^+<k^+$. They must run over $\Lambda<p^+<k^+-\Lambda$, both the plus-prescribed terms and the regular ones:

$$\int_0^{k^+}\frac{dp^+}{2\pi}\to\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi},\qquad\int_0^{k^+}\frac{dq^+}{2\pi}\to\int_\Lambda^{k^+-\Lambda}\frac{dq^+}{2\pi},\qquad\int_{k^+}^{0}\frac{dq^+}{2\pi}\to\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}. \tag{7.6}$$

Table 4 shows the difference.

**(d) Row V, 2b UV.** (1.27) is exact. (1.28) drops $\frac{4\Lambda}{k^+}-\frac{\Lambda^2}{k^{+2}}+\frac{2\Lambda^3}{3k^{+3}}$ but keeps $2\log(1-\frac{\Lambda}{k^+})$, which is of the same order. Keep (1.27).

# 8 G6: notation

- (1.1), (1.2): $\log\Big[1-\frac{k^+}{\vee}\Big]^2$ means $\log^2(1-\frac{k^+}{\vee})$ (X6), not $2\log(1-\frac{k^+}{\vee})$.
- (1.3), (1.8): add $\int_{x'}$. Write the colour factor as $U^{ah}(w)f^{hbc}\{\dots\}-f^{abc}U^{bd}(x)U^{ce}(y)\{\dots\}$, with no $f^{abc}$ in the prefactor (G2).
- Before (1.23): $f^{abc}f^{def}U^{be}(z)U^{cf}(z)=N_c\,U^{ad}(z)$. It holds for any adjoint $U$.
- (1.18), (1.21)–(1.25): after the $w$ integration the phase $e^{-ik\cdot(z-x)p^+/k^+}$ depends on $p^+$ and belongs inside $\int dp^+$.
- (1.18), (1.19): $\int_{x,z,y,x',y'w'}\to\int_{x,z,y,x',y',w'}$.

# 9 The $1/\epsilon$ bracket of Row II (and fix F6)

In $d_\perp=2-\epsilon$, with every transverse integral carrying $(\mu^2e^{\gamma_E}/4\pi)^{\epsilon/2}$ ($\overline{\text{MS}}$), we have $\int d^dz\,K(w,w,z)=0$ (scaleless) and

$$\int d^dz\,K(x,w,z)=\pi^{2-\frac{d}{2}}\Gamma\Big(\frac{d}{2}-1\Big)\big[(x-w)^2\big]^{1-\frac{d}{2}}\Big(\frac{\mu^2e^{\gamma_E}}{4\pi}\Big)^{\epsilon/2}=\pi\Big[-\frac{2}{\epsilon}-2\gamma_E-\log\frac{(x-w)^2\mu^2}{4}\Big]+O(\epsilon). \tag{9.1}$$

So your bracket is exactly $\frac{1}{\pi}\int_z[K(x,w,z)-K(w,w,z)]$ (`sympy`: difference 0). Two consequences:

- It is the Fourier transform of the momentum-space $[-\frac{2}{\epsilon}+\log\frac{k^2}{\mu^2}]$ by your (i8).
- It is the same scheme in which Row V part 2b gives $[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2}{4}]$ (`Part2b_MSbar.pdf`).

Fix F6 of the two-$\rho$ note asked for exactly this form. The different constant quoted there, $\frac{1}{\bar\epsilon}-\log(4\pi^2\mu^2(x-w)^2)$, belongs to $d=2-2\epsilon$ with $\mu^{2\epsilon}$. In the scheme above nothing changes.

## 10 The four-$\rho$ parts of Rows III and IV

These are not in `Group_I.tex`. From the $\{\rho,\rho\}$ entry of $\mathbb{B}_3$ and the two-$\rho$ part of $\mathbb{B}_2$:

$$\begin{aligned}\text{III:}\quad &\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_\Lambda^\vee\frac{dp^+}{2\pi}\frac{1}{p^+}\int e^{-ik\cdot(w'-w)}\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y'-z)\cdot\mathcal{K}(y-z)\\ &\times\big[U(x')-U(w')\big]^{ab'}\rho^{b'}(x')\,\{\bar\rho^a(x),\bar\rho^b(y')\}\,\big[U(z)-U(y)\big]^{bc}\rho^c(y),\end{aligned}\tag{10.1}$$

$$\begin{aligned}\text{IV:}\quad &-\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\frac{1}{p^+}\int e^{-ik\cdot(w'-w)}\mathcal{K}(x'-w')\cdot\mathcal{K}(y'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\\ &\times\big[U(x')-U(w')\big]^{ab'}\rho^{b'}(x')\Big[U^{ac}(w)U^{bd}(z)\,\bar\rho^b(y)\{\rho^c(y'),\rho^d(x)\}-\bar\rho^b(y)\{\bar\rho^a(y'),\bar\rho^b(x)\}\Big],\end{aligned}\tag{10.2}$$

with $\bar\rho(x)=U(x)\rho(x)$, integrated over all transverse points. The $p^+$ integrals give $\frac{1}{2\pi}\log\frac{\vee}{\Lambda}$ and $\frac{1}{2\pi}\log\frac{\vee-k^+}{\Lambda}$. Both agree with first principles (rel. $2\times10^{-14}$), for a general $U$ as well.

## 11 Checks

Scripts: `fp1.py` (first principles), `user1.py` (your equations as written), `wf3.py` ($\mathbb{B}_3$), `checks1.py` (all comparisons, written to `checks1.json`). Random transverse points, random SU(3) adjoint Wilson lines (symmetric, and general) and random charges are used throughout.

Table 2: Pointwise and integrated comparisons. "rel." is the largest relative difference.

| check | result |
|---|---|
| $\mathbb{B}_3$ entries vs. light-cone wave functions (Table 1) | ratio $8\pi^3$, except $-8\pi^3$ for the non-inst. $p^+>k^+$ entry |
| $\frac{1}{\pi}\int_z[K(x,w,z)-K(w,w,z)]$ in $d_\perp=2-\epsilon$, $\overline{\text{MS}}$, minus the Row II bracket | 0 (`sympy`) |
| Row II part 1 (1.1) vs. first principles (any bracket; symmetric and general $U$) | rel. $6\times10^{-16}$ |
| Row II rewriting (1.2) | $4\times10^{-16}$ |
| Row II part 2: (1.3) as written minus `Calculation_main.tex` bracket minus $k^i\log\frac{\vee}{k^+}$ | rel. $2\times10^{-15}$ |
| Row II part 2: split 2a + 2b vs. full bracket (random $\mathbf{k},\mathbf{p}$, $k^+$, $\Lambda$, $\vee$) | rel. $2\times10^{-15}$ |
| Row II part 2: prefactor and colour vs. first principles | rel. $2\times10^{-14}$ |
| Row II part 2: exact bracket (X4, X7) $=(E+2k+D)[\dots]+R_{\text{exact}}$ | rel. $2\times10^{-15}$ |
| Row III (1.10) vs. first principles with $\mathbb{B}_3$ as written (48 points) | rel. $1\times10^{-14}$ |
| Row III (1.10) vs. first principles with E3 | $p^+<k^+$: rel. $1\times10^{-14}$; $p^+>k^+$: differ by $\geq 164\%$ |
| Row III: $N_1,N_2,N_{3a},N_{3b}$ vs. numerical PV integrals, logs as $\log\lvert\cdot\rvert$ (6 configurations) | rel. $3\times10^{-14}$ |
| Row III: the same with complex logarithms, pole inside the range | rel. $1\times10^{-2}$ to $8\times10^{1}$ |
| Row III: green terms with G1 | rel. $4\times10^{-16}$ |
| Row IV (1.15) vs. first principles (symmetric and general $U$) | rel. $5\times10^{-15}$ |
| Row IV (1.16) vs. numerical $\int_\Lambda^{\vee-k^+}$ of (1.15) | rel. $3\times10^{-16}$ |
| Row V part 1 (1.17) vs. first principles | rel. $4\times10^{-15}$ |
| Row V part 2 (1.20) vs. first principles | rel. $1\times10^{-15}$ |
| Four-$\rho$ parts of Rows III and IV (Sec. 10) | rel. $2\times10^{-14}$ |

Table 3: Row III: every $p^+$ integral against the numerical principal-value integral of the line it comes from. $L_1$, $L_2$: instantaneous term for $p^+ > k^+$ and $p^+ < k^+$ (lines 1–3 and 4–6 of $N_1$); $L_3$: the last entry of $\mathbb{B}_3$ (lines 7–8 of $N_1$); $L_4$: non-instantaneous term for $p^+ > k^+$ ($N_2$); $L_5$: non-instantaneous term for $p^+ < k^+$ ($N_{3a} + N_{3b}$). A $*$ marks a pole inside the range. "G1" compares $-N_2$ with the first-principles integral with the corrected $\mathbb{B}_3$.

| configuration | largest relative difference |
|---|---|
| symm. $U$, $k^+ = 1.3$, $\Lambda = 0.07$, $\vee = 9$, $p^+_* = 0.822$ | $L_1$: $9 \times 10^{-16}$, $L^*_2$: $6 \times 10^{-16}$, $L^*_3$: $7 \times 10^{-15}$, $L_4$: $7 \times 10^{-16}$, $L^*_5$: $4 \times 10^{-16}$, G1: $7 \times 10^{-16}$ |
| symm. $U$, $k^+ = 1.3$, $\Lambda = 0.07$, $\vee = 9$, $p^+_* = 91.6$ | $L_1$: $6 \times 10^{-16}$, $L_2$: $3 \times 10^{-16}$, $L_3$: $1 \times 10^{-15}$, $L_4$: $9 \times 10^{-16}$, $L_5$: $7 \times 10^{-16}$, G1: $3 \times 10^{-16}$ |
| symm. $U$, $k^+ = 1.3$, $\Lambda = 0.07$, $\vee = 9$, $p^+_* = 0.246$ | $L_1$: $8 \times 10^{-16}$, $L^*_2$: $7 \times 10^{-16}$, $L^*_3$: $5 \times 10^{-16}$, $L_4$: $9 \times 10^{-16}$, $L^*_5$: $8 \times 10^{-16}$, G1: $5 \times 10^{-16}$ |
| symm. $U$, $k^+ = 1.3$, $\Lambda = 0.07$, $\vee = 9$, $p^+_* = 0.135$ | $L_1$: $8 \times 10^{-16}$, $L^*_2$: $2 \times 10^{-15}$, $L^*_3$: $7 \times 10^{-16}$, $L_4$: $3 \times 10^{-15}$, $L^*_5$: $2 \times 10^{-17}$, G1: $6 \times 10^{-15}$ |
| general $U$, $k^+ = 0.8$, $\Lambda = 0.20$, $\vee = 3$, $p^+_* = 0.506$ | $L_1$: $3 \times 10^{-16}$, $L^*_2$: $4 \times 10^{-16}$, $L^*_3$: $6 \times 10^{-16}$, $L_4$: $2 \times 10^{-15}$, $L^*_5$: $5 \times 10^{-16}$, G1: $1 \times 10^{-15}$ |
| general $U$, $k^+ = 2.0$, $\Lambda = 0.01$, $\vee = 40$, $p^+_* = 8.53$ | $L^*_1$: $2 \times 10^{-14}$, $L_2$: $1 \times 10^{-14}$, $L^*_3$: $4 \times 10^{-18}$, $L^*_4$: $3 \times 10^{-14}$, $L_5$: $1 \times 10^{-14}$, G1: $2 \times 10^{-14}$ |

| seed | $\Lambda/k^+$ | part 1, $\int_0^{k^+}$ | part 1, $\int_\Lambda^{k^+-\Lambda}$ | 2a, $\int_0^{k^+}$ / $\int_\Lambda^{k^+-\Lambda}$ |
|---|---|---|---|---|
| 1 | 0.077 | $6 \times 10^{-3}$ | $2 \times 10^{-16}$ | $9 \times 10^{-2}$ / 0 |
| 1 | 0.008 | $3 \times 10^{-4}$ | $5 \times 10^{-16}$ | $4 \times 10^{-3}$ / $3 \times 10^{-16}$ |
| 1 | 0.001 | $2 \times 10^{-5}$ | $4 \times 10^{-15}$ | $2 \times 10^{-4}$ / $4 \times 10^{-15}$ |
| 2 | 0.077 | $3 \times 10^{-2}$ | $2 \times 10^{-17}$ | $9 \times 10^{-2}$ / $2 \times 10^{-16}$ |
| 2 | 0.008 | $2 \times 10^{-3}$ | 0 | $4 \times 10^{-3}$ / $3 \times 10^{-17}$ |
| 2 | 0.001 | $1 \times 10^{-4}$ | $4 \times 10^{-17}$ | $2 \times 10^{-4}$ / $4 \times 10^{-15}$ |
| 3 | 0.077 | $5 \times 10^{-1}$ | $8 \times 10^{-16}$ | $4 \times 10^{-1}$ / $6 \times 10^{-16}$ |
| 3 | 0.008 | $6 \times 10^{-2}$ | $2 \times 10^{-15}$ | $5 \times 10^{-2}$ / $2 \times 10^{-15}$ |
| 3 | 0.001 | $2 \times 10^{-3}$ | $1 \times 10^{-14}$ | $1 \times 10^{-3}$ / $2 \times 10^{-14}$ |

Table 4: Row V: plus-prescribed forms (1.19), (1.26) against the exact $p^+$ integral of (1.18), (1.24) over $\Lambda < p^+ < k^+ - \Lambda$ (relative difference; $k^+ = 1.3$).